

Internet Engineering Task Force  
Internet-Draft  
Intended status: Standards Track  
Expires: 26 October 2026

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24 April 2026

Extensible Provisioning Protocol (EPP) Transport over QUIC  
draft-ietf-regext-epp-quic-06

Abstract

This document describes how an Extensible Provisioning Protocol (EPP) session is mapped onto a QUIC connection. EPP over QUIC (EoQ) leverages the performance and security features of the QUIC protocol as an EPP transport.

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**Commenté [MB1]:** -- Things are more subtle than that. Please check for example <https://datatracker.ietf.org/meeting/118/materials/slides-118-maprg-sustained-throughput-performance-of-quic-implementations> Or <https://ieeexplore.ieee.org/document/10060785>

There are many studies out there.

-- Do we have a comparative study under typical EPP deployment conditions to benchmark the options out there?

**Commenté [MB2]:** -- This mainly based on TLS. That's not specific to QUIC.

**Commenté [MB3]:** -- What are expectations we have on the sessions (short/long lived)? Do we have experience with long-lived sessions for example?

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1. Introduction

This document describes how the Extensible Provisioning Protocol (EPP) [RFC5730] is mapped onto the QUIC transport [RFC9000]. QUIC is a ~~network~~ protocol that is based on UDP and incorporates native encryption support using TLS [RFC9001]. Though based on UDP, QUIC provides connection semantics like other stateful protocols. This document discusses how EPP implementations can work with this and other features of QUIC while preserving the core EPP semantics.

Commenté [MB4]: Avoid confusing with network layer (IP).

2. Terminology

~~Commonly used terms in this document are described below. This document makes use of the following terms:~~

Commenté [MB5]: Postion as a section to ease referencing (including by other documents).

- EoQ: The acronym used for the EPP over QUIC transport that defines the use of QUIC as an EPP transport following the considerations in Section 2.1 of [RFC5730].
- EoQ connection: Is a client-initiated bidirectional QUIC stream established on a QUIC connection using the "EoQ" Application-Layer Protocol Negotiation (ALPN) [RFC7301] value. The EoQ connection maps to the client-server connection defined in Section 2.1 of [RFC5730], where the server returns an EPP ~~<greeting>~~greeting. A ~~single~~ QUIC connection supports many EoQ connections.
- EoQ session: Is an authenticated EoQ connection, which occurs after a successful EPP <login>.
- EoQ Connection Start Packet: Used by ~~the clients~~ to complete the

creation of an EoQ connection by signaling the server to create the QUIC stream and return the EPP ~~<greeting> greeting~~ needed for an EPP connection. Section 6 formally defines the EoQ Connection Start Packet.

## 2.1 Conventions Used in This Document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

## 3. Session Management

Mapping EPP session management facilities onto the QUIC service is accomplished with a combination of a QUIC connection with the "EoQ" ALPN [RFC7301] value and client-initiated, bidirectional QUIC streams. QUIC supports four stream types [Section 2.1 of [RFC9000]], but EoQ only supports the client-initiated, bidirectional stream type.

An EPP session first requires creation of a QUIC connection between two peers, one that initiates the connection request and one that responds to the connection request. The initiating peer is called the "client", and the responding peer is called the "server". By default, An-an EPP server MUST listen for QUIC connection requests on a ~~standard well-known~~ UDP port number assigned by IANA (see Section 8.1), unless there is a mutual agreement to use another port number.

A successfully established QUIC connection is ~~automatically~~ secured by the native TLS support that QUIC provides using the "EoQ" ALPN [RFC7301] value.

Once the QUIC connection is established, the EPP client MUST then create a bidirectional QUIC stream by sending the EoQ Connection Start Packet (Section 6). [RFC9000] states that "streams are created by sending data". ~~Once-If~~ the EPP server accepts the QUIC stream, it reads the EoQ Connection Start Packet (Section 6) and returns an EPP <greeting> to the client on the same QUIC stream. After reading the EPP <greeting> message, ~~and absent processing errors~~, the EPP client sends EPP commands and receives EPP responses on the same stream.

A QUIC stream corresponds to an EPP connection, which is referred to as a-an EoQ connection. An authenticated QUIC stream, via a successful EPP <login>, corresponds to an EPP session, which is referred to as a-an EoQ session.

An EPP session is normally ended by the client issuing an EPP <logout> command. A server receiving an EPP <logout> command MUST end the EPP session and close the QUIC stream. A client MAY end an EoQ session by closing the QUIC stream and the server MUST end the

**Commenté [MB6]:** Add a pointer whether the new ALPN is defined, otherwise this can be misinterpreted as that alpn is from 7301

**Commenté [MB7]:** Help readers find where to look at.

**Commenté [MB8]:** Move to a new para for better flow.

**Commenté [MB9]:** EoQ?

EPP does not require QUIC :-)

**Commenté [MB10]:** Wouldn't be possible to run the server over another port number?

I would reword to «absent local policy, ...»

**Commenté [MB11]:** -- This assumes that the server has will always accept the connection.

From an operational standpoint, a server may have a policy to rate-limit connections. This MUST is thus to be revisited.

**Commenté [MB12]:** As it can reject it.

**Commenté [MB13]:** Move to a new para for better flow.

**Commenté [MB14]:** -- Did we consider a scheme where each command/response is mapped to a stream?

EoQ session by closing the QUIC stream.

EoQ connections are established as described in the QUIC transport specification [RFC9000]. During connection establishment, EoQ support is indicated using the "EoQ" ALPN ~~[RFC7301]~~ value in the ~~cryptographic crypto~~ handshake.

A single QUIC connection may allow multiple QUIC streams. This means that a single QUIC connection may support multiple EoQ sessions. A server MAY limit the life span of an established EoQ session. EoQ sessions that are inactive for more than a server-defined period MAY be ended by a the server closing the QUIC stream. A server MAY close

EoQ

sessions that have been open and active for longer than a server-defined limit. Once the last QUIC stream for a QUIC connection is closed, the server MAY end the QUIC connection immediately.

#### 4. Message Exchange

Except for the EPP server ~~<greeting>~~greeting, EPP messages are initiated by the

EPP client in the form of EPP commands. An EPP server MUST return an EPP response to an EPP command on the same QUIC stream that carried the command. If the QUIC stream is closed after a server receives and successfully processes a command but before the response can be returned to the client, the server MAY attempt to undo the effects of the command to ensure a consistent state between the client and the server. **EPP commands are idempotent, so processing a command more than once produces the same net effect on the repository as successfully processing the command once.**

An EPP client streams EPP commands to an EPP server on an established QUIC stream. A client MAY establish multiple QUIC streams to support multiple EoQ sessions with each EoQ session mapped to a single QUIC stream. A server SHOULD limit a client to a maximum number of QUIC streams per QUIC connection based on server capabilities and operational load. ~~Absent such limit, the server may be subject to overload that would exhaust its resources.~~

EPP describes client-server interaction as a command-response exchange where the client sends one command to the server and the server returns one response to the client.

Each EPP data unit MUST contain a single EPP message. Commands MUST be processed independently.

A server SHOULD impose a limit on the amount of time required for a client to issue a well-formed EPP command to reduce the risk associated with a resource exhaustion attack. ~~Absent local policy, A-a~~ server SHOULD end an EoQ session and close the QUIC stream if a well-formed command is not received within the time limit.

A general state machine for an EPP server is described in Section 2 of [RFC5730]. A general client-server message exchange using QUIC transport is illustrated in Figure 1. It shows the exchange over a single QUIC stream of a QUIC connection. Many QUIC streams may open and close during the life of a QUIC connection.

**Commenté [MB15]:** (OPS): Do we need to have specific error messages to ease diagnostic? See, for example, <https://datatracker.ietf.org/doc/html/rfc9250#name-dog-error-codes>

**Commenté [MB16]:** Be consistent with RFC9000

**Commenté [MB17]:** Consider adding a discussion about required configuration paramters in an Operational Considerations section to cover these.

**Commenté [MB18]:** Consider adding an operational considerations to cover logging.

**Commenté [MB19]:** This is better moved to an Operational Considerations section.

**Commenté [MB20]:** To explain the effect of not following the SHOULD.

**Commenté [MB21]:** I guess failing to adhere to this MUST will lead to connection closure. How to signal the error to the peer?

**Commenté [MB22]:** This text also better in an OPS Cons section to better emphasis what an operator has to do.

**Commenté [MB23]:** Otherwise, we need to explain why this is not a MUST

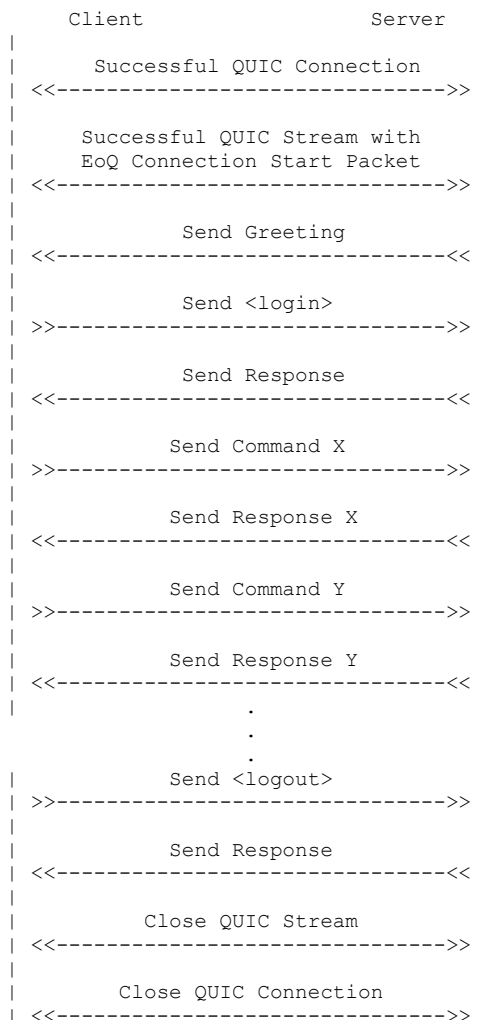


Figure 1: **Example of Successful** QUIC Client-Server

Message Exchange

The EPP server MUST follow the "EPP Server State Machine" procedure described in [RFC5730].

## 5. Data Unit Format

The EPP data unit contains two fields: a 32-bit header that describes the total length of the data unit, and the EPP XML instance. The length of the EPP XML instance is determined by subtracting four octets from the total length of the data unit. A receiver must

**Commenté [MB24]:** As the flow may be different if an error is encountered.

successfully read that many octets to retrieve the complete EPP XML instance before processing the EPP message.

The EPP Data Unit Format is depicted in Figure 2 (one tick mark represents one bit position).

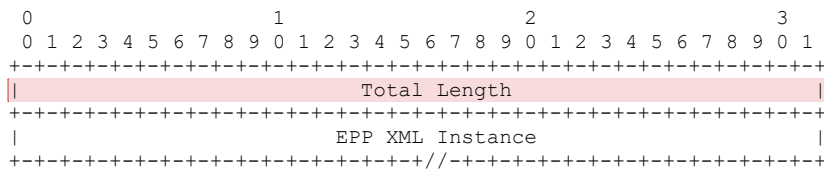


Figure 2: EPP Data Unit Format

Commenté [MB25]: -- size and fragmentation

Please add a discussion under OPS Cons about size/MTU and fragmentation. May refer to rfc9000#section-14

The description of the fields shown in Figure 2 is as follows:

Total Length (32 bits): The total length of the EPP data unit measured in octets in network (big endian) byte order. The octets contained in this field MUST be included in the total length calculation.

EPP XML Instance (variable length): The EPP XML instance carried in the data unit.

#### 6. EoQ Connection Start Packet

The EoQ Connection Start Packet is written by the client after creating a QUIC stream to signal to the server to create the QUIC stream. Absent processing errors or local policy, The-the server accepts the QUIC stream, reads the EoQ Connection Start Packet, and returns the EPP <greeting> to the client on same QUIC stream.

Commenté [MB26]: Do we need to negotiate idle timeout? See for example the use in 4.4 of RFC9250

Commenté [MB27]: Otherwise, the following steps may not be applicable systematically.

The EoQ Connection Start Packet follows the Data Unit Format (Section 5) with two fields: a 32-bit header that describes the total length of the data unit, and the constant value of "EoQ Connection Start" instead of an "EPP XML Instance". The length of the-a valid data unit will-MUST be 24 octets that includes 4 octets for the Total Length and 20 octets for the "EoQ Connection Start" constant value.

#### 7. Transport Considerations

Section 2.1 of [RFC5730] describes considerations to be addressed by protocol transport mappings. This document addresses each of those considerations using a combination of features of the QUIC protocol itself-and features of this document-as discussed below:-:

- \* Command Order: QUIC guarantees ordered processing of data within each stream. Section 2 of [RFC9000] describes streams in detail.
- \* Session Mapping: EPP session management utilizes QUIC streams and is described in Section 3

\* Stateful Nature: QUIC supports stateful communications between endpoints via ~~connection~~ Connection IDs and long-lived streams within each connection. Sections 2 and 5 of [RFC9000] describe these features, respectively.

Commenté [MB28]: Be consistent with RFC9000

\* Frame Data Units: EoQ uses the packet framing defined in Section 5.

\* Congestion Avoidance: QUIC provides various mechanisms to help achieve congestion avoidance. [RFC9002] describes these mechanisms in detail.

\* Reliability: QUIC uses message acknowledgement, packet retransmission, and other features to ensure reliability. ~~Part~~ packetization of Section 13 [RFC9000] describes these features in detail.

Commenté [MB29]: Better to have a direct link

\* Pipelining: Pipelining is allowed in EoQ. QUIC streams support sending multiple frames without waiting for responses from the other peer. This does not change the basic single command, single response operating mode of the core EPP ~~protocol~~.

Commenté [MB30]: Redundant with «P» of EPP

Commands MUST be processed independently and in the same order as sent from the client.

Batch-oriented processing (combining multiple EPP commands in a single data unit) is not permitted. ~~Each EPP data unit MUST~~ must contain a single EPP message.

Commenté [MB31]: This is already stated. No need to use normative language here.

An EPP x5zz "Connection Management" error response, defined in Section 3 of [RFC5730], of a well-formed EPP client packet results in the server closing the EoQ connection after returning the error response. A malformed EPP client packet results in the server closing the EoQ connection without providing an error response. All subsequent EPP commands sent on the EoQ connection will not be processed.

## 8. IANA Considerations

RFC Editor Note: Please replace all occurrences of XXXX with the RFC number to be assigned to this document.

### 8.1. Registration of an EoQ Identification String

This document creates a new registration for the identification of EoQ in the "TLS Application-Layer Protocol Negotiation (ALPN) Protocol IDs" registry ~~{RFC7301}~~ under "Transport Layer Security (TLS) Extensions" registry group available at <https://www.iana.org/assignments/tls-extensiontype-values/>.

\* Protocol: EoQ

\* Identification Sequence: 0x45 0x6F 0x51 ("EoQ")

\* Reference: ~~This document~~ RFC XXXX

## 8.2. Registration of Port Number

The "Service Name and Transport Protocol Port Number Registry" (<https://www.iana.org/assignments/service-names-port-numbers/>) contains an entry for EPP UDP/700 ~~based on [RFC6335]~~. However, no known implementations of EPP over UDP exist.

~~The~~This document requests IANA to update that ~~entry will be so that it is~~ reassigned to EPP and add a reference to this ~~draft~~ document.

- \* Service Name: epp
- \* Port Number: 700
- \* Transport Protocol(s): UDP
- \* Assignee: IESG
- \* Contact: IETF Chair
- \* Description: EPP run over QUIC
- \* Reference: ~~This document~~ [RFC5734] RFC XXXX

**Commenté [MB32]:** We need to cover the lack of ops impact in the OPS Considerations.

## 9. Implementation Status

Note to RFC Editor: Please remove this section and the reference to RFC 7942 [RFC7942] before publication.

This section records the status of known implementations of the protocol defined by this specification at the time of posting of this Internet-Draft, and is based on a proposal described in RFC 7942 [RFC7942]. The description of implementations in this section is intended to assist the IETF in its decision processes in progressing drafts to RFCs. Please note that the listing of any individual implementation here does not imply endorsement by the IETF. Furthermore, no effort has been spent to verify the information presented here that was supplied by IETF contributors. This is not intended as, and must not be construed to be, a catalog of available implementations or their features. Readers are advised to note that other implementations may exist.

According to RFC 7942 [RFC7942], "this will allow reviewers and working groups to assign due consideration to documents that have the benefit of running code, which may serve as evidence of valuable experimentation and feedback that have made the implemented protocols more mature. It is up to the individual working groups to use this information as they see fit".

### 9.1. Verisign EPP SDK

Organization: Verisign Inc.

Name: Verisign EPP SDK

**Commenté [MB33]:** Check if we need to remove that reference.

Make sure we have the request clear for IANA.



Description: The Verisign EPP SDK includes both a full client implementation and a full server stub implementation of this specification.

Level of maturity: Development

Coverage: All aspects of the protocol are implemented with QUIC V1.

Licensing: GNU Lesser General Public License

Contact: [jgould@verisign.com](mailto:jgould@verisign.com)

URL: <https://www.verisign.com/resources/registrar-resources/epp-sdk/>

## X. Operational Considerations

### X.1. Client's Fall Back and Management of Multiple Transport

See for example 5.2 of RFC 9250

### X.2 Port Reuse

Although [RFC5734] does only a request for TCP, the companion UDP number was also allocated. That practice was prior to [RFC6335] when TCP and UDP port numbers were simultaneously assigned when either was requested.

Section 8.2 updates EPP UDP/700 allocation to be used for EoQ. This update does not introduce any operational issues given that there is no known implementations of EPP over UDP exist.

### X.3. QUIC Support Announcement and Discovery

Advisory means to know that a server supports QUIC

### X.4. Configuration Parameters

Cover the various configuration parameters/limits

### X.5 Diagnostic and Troubleshooting

Logging, specific errors, etc.

### X.6. Address Validation

See for example 5.3 of RFC 9250

### X.7. Authentication Considerations

### X.8. 0-RTT and Session Resumption

See for example 5.5.3 of RFC 9250

### X.9. MTU and Fragmentation

## 10. Security Considerations

EPP over QUIC provides ~~the~~-similar security protection as with EPP over TCP with

TLS. ~~Some related~~Readers should refer to the ~~security issues~~  
~~considerations have been~~ discussed in Section 8 of [RFC5734]  
And Section 21 [RFC9000].

EOQ servers run the risk of a resource exhaustion attack by allowing  
the creation of unlimited QUIC streams per QUIC connection. Servers  
MUST support guards to prevent overload. Specifically, servers  
~~SHOULD have to~~ limit a client to a maximum number of QUIC streams per  
QUIC  
connection based on server capabilities and operational load.  
Likewise, servers are RECOMMENDED limit the number of QUIC connections  
bound to the same client to ensure usage fairness among clients invoking  
the EPP service.

## 11. Acknowledgements

The authors wish to thank the following persons for their feedback  
and suggestions: Scott Hollenbeck, Lucas Pardue, and Martin  
~~Thompson~~Thomson.

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Commenté [MB34]: Please ACK RFC5734 as the text is  
inspired from that RFC.

Commenté [MB35]: Should be informative.

Will be removed anyway 😊

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